Ian Vasconcellos Caldas

Curriculum vitae
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Department of Computational Biology
Cornell University
101 Biotechnology Building
Ithaca, NY 14850
United States

ivc2@cornell.edu
+1 (607) 280-9461
github.com/ianvcaldas
twitter.com/ianvcaldas

Education

2016–present	PhD candidate in Computational Biology at Cornell University. Advisors: Dr. Andrew G. Clark and Dr. Philipp W. Messer.
2013–2015	M.Sc. in Genetics at Federal University of Rio de Janeiro. Advidor: Dr. Carlos G. Schrago. Thesis: Strategies to estimate with precision the divergence time of placental mammals using the morphological clock.
2009–2013	B.Sc. in Genetics at Federal University of Rio de Janeiro. Advisor: Dr. Carlos G. Schrago. Thesis: Performance of Bayesian skyline plots to estimate the demographic history of structured populations.

Publications

Caldas, I. V. & Schrago, C. G. (2019). Data partitioning and correction for ascertainment bias reduce the uncertainty of placental mammal divergence times inferred from the morphological clock. *Ecology and Evolution*, *9*(4), 2255–2262. doi:10.1002/ece3.4921

Duneau, D., Sun, H., Revah, J., San Miguel, K., Kunerth, H. D., **Caldas, I. V.**, … Buchon, N. (2018). Signatures of insecticide selection in the genome of drosophila melanogaster. *G3: Genes, Genomes, Genetics*, *8*(11), 3469–3480. doi:10.1534/g3.118.200537

Wei, K. H.-C., Lower, S. E., **Caldas, I. V.**, Sless, T. J. S., Barbash, D. A., & Clark, A. G. (2018). Variable rates of simple satellite gains across the drosophila phylogeny. *Molecular Biology and Evolution*, *35*(4), 925–941. doi:10.1093/molbev/msy005

Flynn, J. M., **Caldas, I. V.**, Cristescu, M. E., & Clark, A. G. (2017). Selection constrains high rates of tandem repetitive dna mutation in daphnia pulex. *Genetics*, *207*(2), 697–710. doi:10.1534/genetics.117.300146

Selected conference presentations

2019 Evolution.
Poster: Inferring parameters of selective sweeps through supervised learning.

2018 Population, Evolutionary and Quantitative Genetics Conference.
Poster: Inference of sweep parameters through supervised learning.

2017 Society for Molecular Biology and Evolution.
Poster: Tandem short repetitive DNA recapitulates the evolutionary history in *Drosophila*.

2015 Evolution.
Poster: Data partitioning improves the morphological clock of placental mammals.

Awards

2016 Presidential Life Sciences Scholarship, Cornell University.

2013 Master's Research Scholarship, National Council for Scientific and Technological Development of Brazil (CNPq).

Professional society memberships

2015–present Society for the Study of Evolution (SSE).

2015–present Society for Molecular Biology and Evolution (SMBE).

Further training

Scientific workshops

2018 Search for Selection, The National Institute for Mathematical and Biological Synthesis.

2017 High Performance Computing on Stampede 2, Cornell University.

2015 Applied Bayesian Modeling, Osvaldo Cruz Foundation.

2014 Paleobiological and Phylogenetic Approaches to Macroevolution, National Evolutionary Synthesis Center.

Leadership training

2019 Future Professors Institute: Advancing Diversity in the Academy, Cornell University.

2017 Intergroup Dialogue Project, Cornell University.

Skills

Computer skills

Programming languages	Python, R
Bioinformatics	Unix shell scripting, genomics pipelines, phylogenetic inference
Data analysis	scikit-learn, fastai, tidyverse, data visualization with ggplot2 and matplotlib
Other	LaTeX

Languages

Native	Portuguese
Fluent	English, German